

B.Tech. Degree VI Semester Examination April 2018

ME 15-1604 HEAT AND MASS TRANSFER (2015 Scheme)

(Approved heat and mass transfer data books are permitted)

Time: 3 Hours

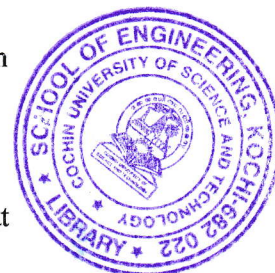
Maximum Marks: 60

PART A

(Answer **ALL** questions)

(10 × 2 = 20)

- I. (a) Explain thermal conductivity. Derive the thermal conduction resistance for a composite plane wall with three different materials in series.
- (b) Derive an expression to find the critical radius of insulation over a cylinder. Give an example where you prefer to have critical radius for the insulation.
- (c) What is the significance of biot number in the lumped system analysis of unsteady conduction? Explain the limit of biot number applicable for a lumped system.
- (d) Draw the boundary layers for a hot vertical plate suspended in atmospheric air. Mark velocity and temperature profiles.
- (e) Explain different stages of pool boiling with pool boiling graph.
- (f) What is meant by a gray body? What is its significance in radiation analysis?
- (g) Explain emissivity. What is the emissivity of sun?
- (h) Explain the significance of NTU for heat exchangers.
- (i) Draw the temperature profile variation for parallel flow/counter flow heat exchangers and condenser.
- (j) Explain the effect of fouling on heat exchanger performance. How does it affect the design of a heat exchanger?



PART B

(4 × 10 = 40)

- II. (a) Consider a cylindrical 0.10 cm diameter copper thermocouple wire originally at 150°C. Determine the time required for the thermocouple to reach equilibrium with 1°C difference with the fluid when this wire is suddenly immersed in water at 40°C (heat transfer coefficient 80 W/m²K) and in air at 40°C (heat transfer coefficient 10 W/m²K). (5)
 - (b) Compare the heat loss from an insulated and an uninsulated copper pipe under the following conditions. The pipe (conductivity 400 W/mK) has an internal diameter of 10 cm and an external diameter of 12 cm. Saturated steam flows inside the pipe at 110°C. The pipe is located in a space at 30°C and the heat transfer coefficient on its outer surface is estimated to be 15 W/m²K. The insulation available to reduce heat losses is 5 cm thick and its conductivity is 0.20 W/mK. (5)
- OR**
- III. (a) A steel pipe line ($k = 50$ W/mK) of ID 100 mm and OD 110 mm is to be covered with two layers of insulation each having a thickness of 50 mm. The thermal conductivity of the first insulation material is 0.06 W/mK and that of the second is 0.12 W/mK. Calculate the loss of heat per metre length of pipe and the interface temperature between the two layers of insulation when the temperature of the inside tube surface is 250°C and that of the outside surface of the insulation is 50°C. (5)

(P.T.O.)

- (b) To increase the heat dissipation from a 2.5 cm OD tube, circumferential fins made of aluminium (conductivity 200 W/mK) are soldered to the outer surface. The fins are 0.1 cm thick and have an outer diameter of 5.5 cm. The tube temperature is 100°C, the environmental temperature is 25°C, and the heat transfer coefficient between the fins and the environment is 65 W/m²K. There are 100 fins per meter length of tube. Calculate the rate of heat loss with fins and without fins (assume same heat transfer coefficient) per meter length of the tube. (5)
- IV. (a) An electronic device is cooled by water flowing through capillary holes drilled in the casing. The temperature of the device casing is constant at 353 K. The capillary holes are 0.3 m long and 2.54×10^{-3} m in diameter. If water enters at a temperature of 333 K and flows at a velocity of 0.2 m/s, calculate the outlet temperature of water. (5)
- (b) Determine the Nusselt number for water flowing at an average velocity of 3 m/s in an annulus formed between a 2.54 cm OD tube and a 34.8 cm ID tube. The water is at 83°C and is being cooled. The temperature of the inner wall is 38°C, and the outer wall of the annulus is insulated. Neglect entrance effects and choose appropriate correlation. (5)
- OR**
- V. (a) Water entering at 10°C is to be heated to 40°C in a tube of 0.02 m ID at a mass flow rate of 0.01 kg/s. The outside of the tube is wrapped with an insulated electric-heating element that produces a uniform flux of 15,000 W/m² over the surface. Neglecting any entrance effects, determine the heat transfer coefficient, the length of pipe needed for a 30°C increase in average water temperature. (5)
- (b) A 2.0 cm diameter horizontal heater is maintained at a surface temperature of 38°C and submerged in water at 27°C. Calculate the free-convection heat loss per unit length of the heater. (5)
- VI. (a) Liquefied oxygen (boiling temperature, -183°C) is to be stored in a spherical container with 0.3 m in diameter. The system is insulated by an evacuated space between the inner sphere and a surrounding 0.46 m ID concentric sphere. Both spheres are made of polished aluminum (emissivity = 0.03), and the temperature of the outer sphere is -1°C. Estimate the rate of heat flow by radiation to the oxygen in the container. (5)
- (b) Two parallel plates 0.5 m by 1.0 m are spaced 0.5 m apart. One plate is maintained at 1000°C and the other at 500 °C. The emissivities of the plates are 0.2 and 0.5, respectively. The plates are located in a very large room, the walls of which are maintained at 27°C. The plates exchange heat with each other and with the room, but only the plate surfaces facing each other are to be considered in the analysis. Find the net transfer to each plate and to the room. (5)
- OR**
- VII. (a) Two very large parallel planes with emissivities 0.3 and 0.8 exchange heat. Find the percentage reduction in heat transfer when a polished-aluminum radiation shield (emissivity 0.04) is placed between them. (5)
- (b) Two large parallel planes having emissivities of 0.3 and 0.5 are maintained at temperatures of 900 K and 400 K, respectively. A radiation shield having an emissivity of 0.05 on both sides is placed between the two planes. Calculate the heat-transfer rate per unit area if the shield were not present, the heat-transfer rate per unit area with the shield present, and the temperature of the shield. (5)

(Contd...3)

- VIII. (a) Determine the heat transfer surface area required for a heat exchanger constructed from a 0.0254 m OD tube to cool 6.93 kg/s of an ethyl alcohol solution (Specific heat $C_p = 3810 \text{ J/kg K}$) from 65.6°C to 39.4°C , using 6.306 kg/s of water available at 10°C ($C_p = 4187 \text{ J/kgK}$). Assume that the overall coefficient of heat transfer based on the outer-tube area is $568 \text{ W/m}^2\text{K}$ and consider each of the following arrangements: (i) parallel-flow (ii) counter-flow. Assume that the alcohol is flowing through the shell and the water is flowing through the tubes. (5)
- (b) Hot oil at 100°C is used to heat air in a shell-and-tube heat exchanger. The oil makes six tube passes and the air makes one shell pass; 2.0 kg/s of air are to be heated from 20°C to 80°C . The specific heat of the oil is $2100 \text{ J/kg}^\circ\text{C}$, and its flow rate is 3.0 kg/s. Calculate the area required for the heat exchanger for $U = 200 \text{ W/m}^2^\circ\text{C}$. (5)

OR

- IX. (a) A shell-and-tube heat exchanger is used as an ammonia condenser with ammonia vapor entering the shell at 50°C as a saturated vapor. Water enters the single-pass tube arrangement at 20°C and the total heat transfer rate required is 200 kW. The overall heat-transfer coefficient is estimated as $1000 \text{ W/m}^2^\circ\text{C}$. Determine the area to achieve a heat exchanger effectiveness of 60 percent with an exit water temperature of 40°C . What percent reduction in heat transfer would result if the water flow is reduced to half while keeping the heat exchanger area and U the same? (6)
- (b) Derive an expression to find the LMTD of a countreflow heat exchanger. (4)

